LAPORAN PRAKTIKUM

POSTTEST 3

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas (B1‘25)

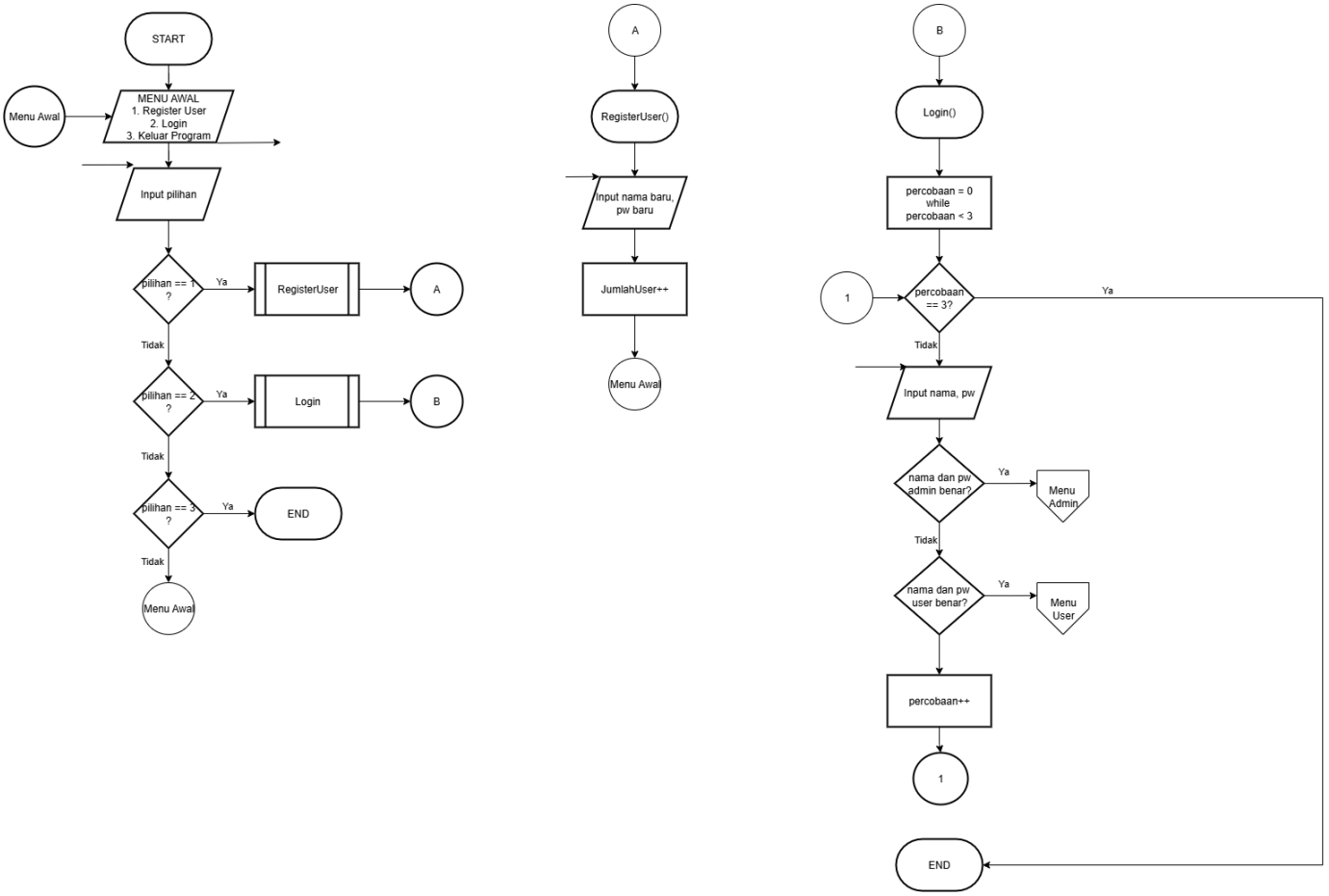
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

2026

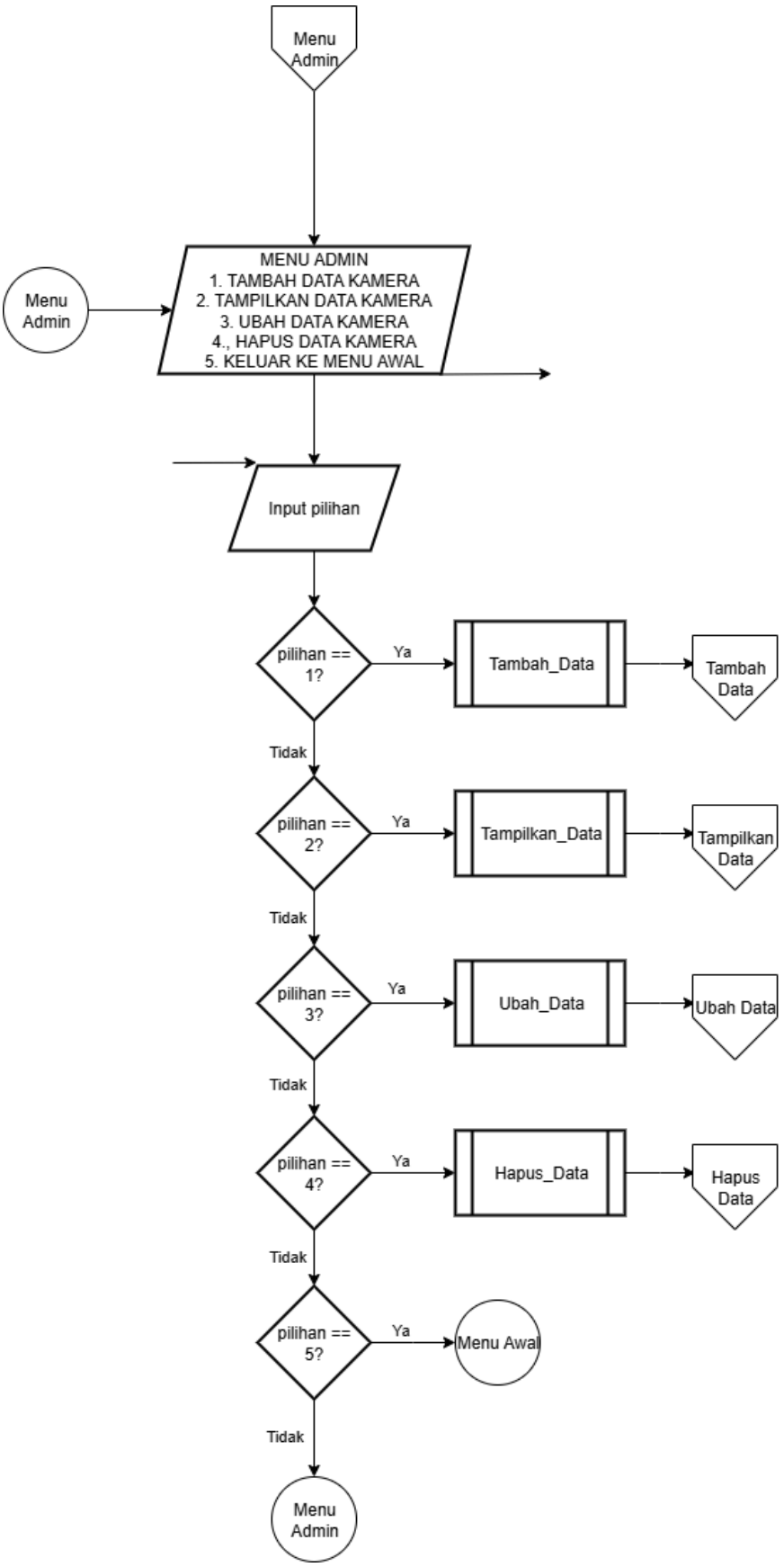
1. Flowchart



Program dimulai. Menu awal memiliki 3 pilihan:

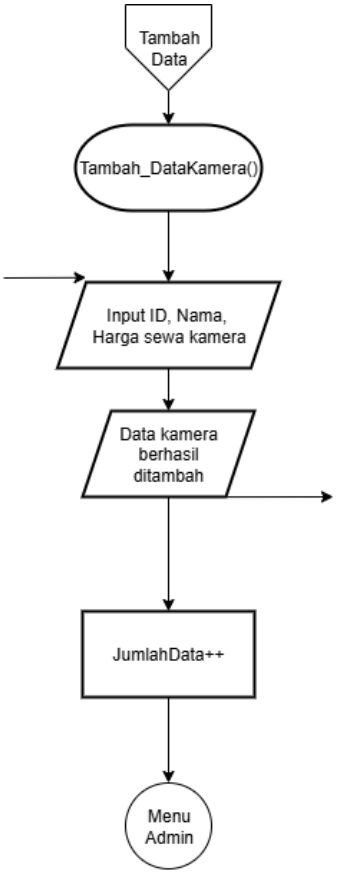
1. Register User
2. Login
3. Keluar Program

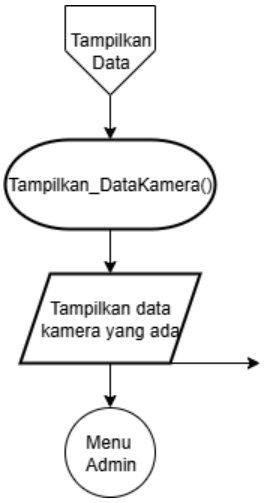
Jika memilih 1, maka pengguna akan diarahkan ke menu register untuk daftar akun baru. Admin diberi nama “TWISTERSIAN” dan passwordnya “123Action”, jadi admin tidak perlu register (bisa langsung login ke program). Jika sudah register, pengguna dapat login jika memilih pilihan 2. Di menu login, pengguna juga memasukkan nama dan password yang sesuai ketika pengguna melakukan register akun. Pengguna diberi percobaan login yang diberi nilai 0 dan ada kondisi yang diberikan selama percobaan login kurang dari 3. Jika salah memasukkan nama atau password, maka percobaan login bertambah sebanyak 1 dan jika sampai 3 kali gagal login, maka program otomatis berhenti. Jika memilih 3, maka program akan berhenti.

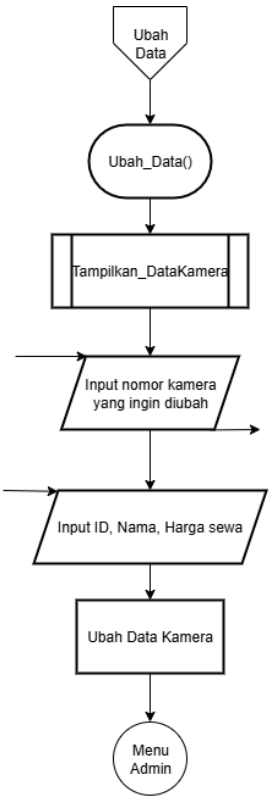

Menu Admin
Menu Admin
MENU ADMIN
1. TAMBAH DATA KAMERA
2. TAMPILKAN DATA KAMERA
3. UBAH DATA KAMERA
4., HAPUS DATA KAMERA
5. KELUAR KE MENU AWAL
Input pilihan
pilihan == 1?
Ya
Tambah_Data
Tambah Data
Tidak
pilihan == 2?
Ya
Tampilkan_Data
Tampilkan Data
Tidak
pilihan == 3?
Ya
Ubah_Data
Ubah Data
Tidak
pilihan == 4?
Ya
Hapus_Data
Hapus Data
Tidak
pilihan == 5?
Ya
Menu Awal
Tidak
Menu Admin

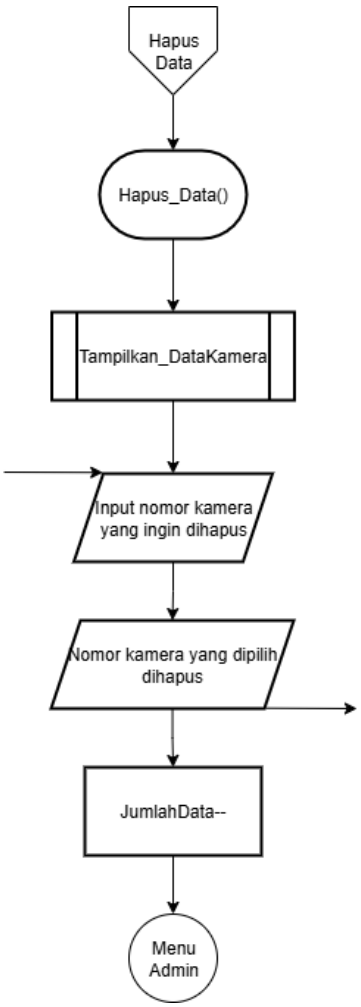
Menu Admin memiliki:

1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Keluar ke Menu Awal





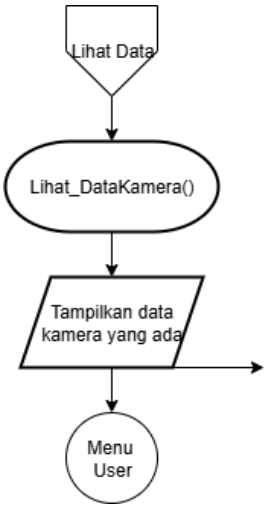

Ubah Data
Ubah_Data()
Tampilkan_DataKamera
Input nomor kamera yang ingin diubah
Input ID, Nama, Harga sewa
Ubah Data Kamera
Menu Admin

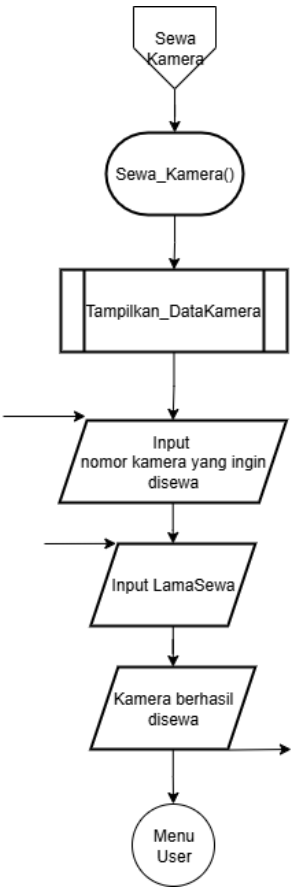


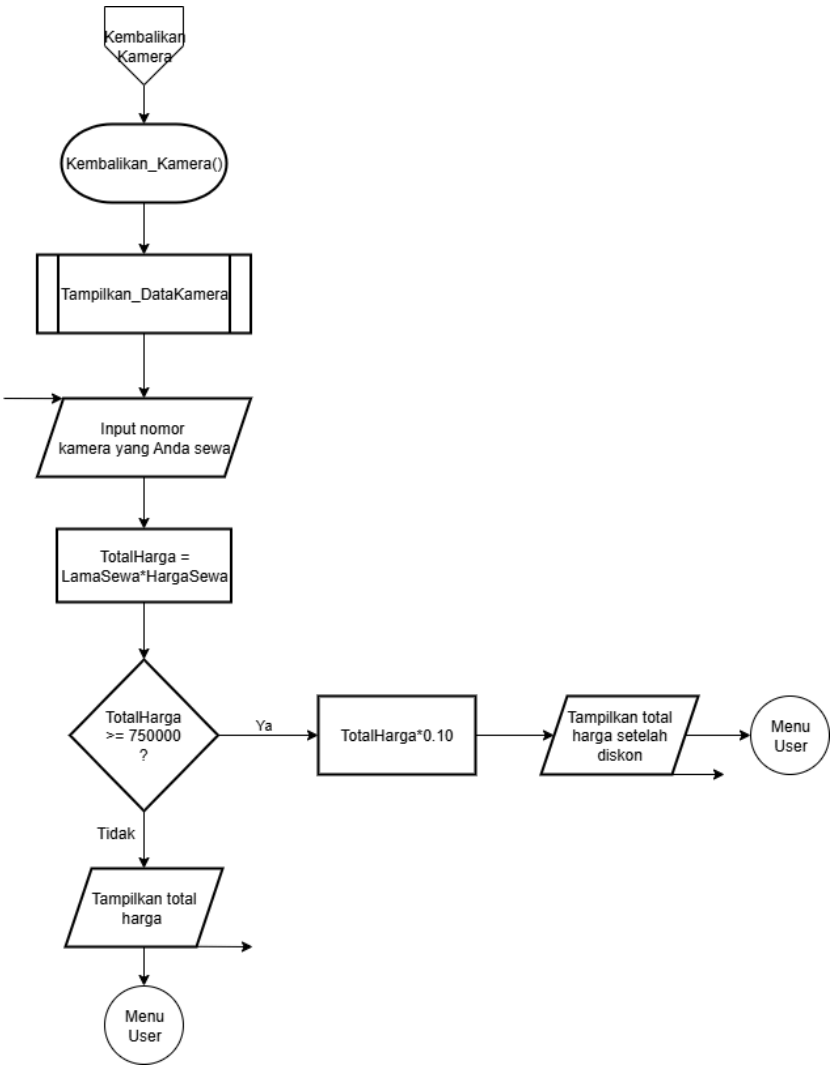
Jika memilih menu nomor 1, maka admin bisa menambah ID kamera, nama kamera, dan harga sewa/hari kamera. Jika memilih menu nomor 2, program akan menampilkan data kamera, baik ID kamera, nama kamera, dan harga sewa per/hari. Jika memilih menu nomor 3, admin dapat mengubah data kamera. Jika memilih menu nomor 4, admin dapat menghapus data kamera. Setelah selesai menggunakan menu 1-4, program akan kembali ke menu admin. Jika memilih menu nomor 5, maka program akan mengarahkan ke menu awal.

Menu User memiliki:

1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera yang Disewa
4. Logout




Sewa Kamera
Sewa_Kamera()
Tampilkan_DataKamera
Input nomor kamera yang ingin disewa
Input LamaSewa
Kamera berhasil disewa
Menu User



Jika memilih nomor 1, user dapat melihat data kamera yang ada. Jika memilih nomor 2, User dapat menyewa kamera yang tersedia. Jika memilih nomor 3, user dapat mengembalikan kamera yang sebelumnya disewa dan program juga akan menghitung total harga sewa berdasarkan harga sewa dan lama sewa. Setelah selesai menggunakan menu 1-3, program akan kembali ke menu user. Jika memilih nomor 4, maka user kembali ke menu awal.

2. Deskripsi Singkat Program

Program Sistem Data Penyewaan Kamera adalah program yang dimana Admin mengelola dan menyimpan data kamera yang terdiri dari ID kamera, nama kamera, harga sewa/hari, dan status kameranya apakah tersedia atau sedang disewa. User dapat menyewa kamera yang sedang tersedia dan dapat mengembalikan kamera sewaannya sesuai dengan kamera yang

disewa dan berapa lama User menyewa kamera itu. Program menggunakan fungsi dan prosedur untuk memecah beberapa menu dari admin dan user.

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct User
{
    string username;
    string password;
    string role;
};

struct StatusKamera
{
    string kondisi;
};

struct Kamera
{
    string ID_Kamera;
    string NamaKamera;
    int HargaSewa;
    int LamaSewa;
    string NamaPenyewa;
    StatusKamera status;
};

void RegisterUser(User user[], int &JumlahUser){
    string username, Password;
    cout << "\n========================\n";
    cout << "       REGISTER USER\n";
    cout << "========================\n";
    cout << "Masukkan nama baru: ";
    getline(cin, user[JumlahUser].username);
```

```
    cout << "Masukkan password baru: ";
    getline(cin, user[JumlahUser].password);
    user[JumlahUser].role = "user";
    JumlahUser++;
    cout << "Registrasi berhasil! Silakan login\n";
}

bool LoginUser(User user[], int JumlahUser, string &InputNama, string
&role_login){
    string InputPassword;
    int percobaan_login = 0;

    while (percobaan_login < 3){
        cout << "\n===================\n";
        cout << "        LOGIN\n";
        cout << "===================\n";
        cout << "Masukkan username: ";
        getline(cin, InputNama);
        cout << "Masukkan password: ";
        getline(cin, InputPassword);

        for (int i = 0; i < JumlahUser; i++){
            if (InputNama == user[i].username && InputPassword ==
user[i].password){
                cout << "Berhasil login! Selamat datang " <<
InputNama << endl;
                role_login = user[i].role;
                return true;
            }
        }
        percobaan_login++;
        cout << "Gagal login! Sisa percobaan: " << (3 -
percobaan_login) << endl;
    }
    cout << "Gagal 3x! Akses ditolak!\n";
    return false;
}

void TambahDataKamera(Kamera data[], int &JumlahData){
    if (JumlahData == 5){
        cout << "Data kamera penuh!";
        return;
```

```
    }
    cout <<
"\n==================================================================
=====================" << endl;
    cout << "                          TAMBAH DATA KAMERA TWISTERS
CAM                                 " << endl;
    cout <<
"==================================================================
====================" << endl;
    cout << "ID Kamera: ";
    getline(cin, data[JumlahData].ID_Kamera);
    cout << "Nama Kamera: ";
    getline(cin, data[JumlahData].NamaKamera);
    cout << "Harga sewa/hari: Rp";
    cin >> data[JumlahData].HargaSewa;
    cin.ignore();
    data[JumlahData].status.kondisi = "Tersedia";
    JumlahData++;
    cout << "Data berhasil ditambahkan";
    cout << endl;
}

void TampilkanDataKamera(Kamera data[], int JumlahData){
    if (JumlahData == 0){
        cout << "Data kamera belum tersedia";
        return;
    }
    cout <<
"\n==================================================================
=====================\n";
    cout << "                           DATA KAMERA TWISTERS CAM\n";
    cout <<
"==================================================================
====================\n";
    cout << left << setw(5) << "No";
    cout << setw(10) << "ID";
    cout << setw(20) << "Nama Kamera";
    cout << setw(20) << "Harga Sewa/hari";
    cout << setw(12) << "Status";
    cout << endl;
    cout <<
"------------------------------------------------------------------
```

```
------------------\n";

    for (int i = 0; i < JumlahData; i++){
        cout << left << setw(5) << i + 1;
        cout << setw(10) << data[i].ID_Kamera;
        cout << setw(20) << data[i].NamaKamera;
        cout << "Rp" << setw(17) << data[i].HargaSewa;
        cout << setw(12) << data[i].status.kondisi << endl;
    }
    cout <<
"=================================================================
====================\n";
}
void UbahDataKamera(Kamera data[], int JumlahData){
    int index;
    if (JumlahData == 0){
        cout << "Data kamera belum tersedia";
        return;
    }
    TampilkanDataKamera(data, JumlahData);
    cout << "Pilih nomor kamera yang ingin diubah: ";
    cin >> index;
    cin.ignore();
    index--;

    if (index >= 0 && index < JumlahData)
    {
        cout << "ID Kamera baru: ";
        getline(cin, data[index].ID_Kamera);
        cout << "Nama Kamera baru: ";
        getline(cin, data[index].NamaKamera);
        cout << "Harga sewa/hari: Rp";
        cin >> data[index].HargaSewa;
        cin.ignore();
        cout << "Status (Tersedia/Disewa): ";
        getline(cin, data[index].status.kondisi);
        cout << "Data kamera berhasil diubah!\n";
    }
    else
    {
        cout << "Nomor tidak valid!";
    }
```

```
}

void HapusDataKamera(Kamera data[], int &JumlahData){
    if (JumlahData == 0){
        cout << "Data kamera tidak tersedia!";
        return;
    }
    int index;
    char konfir_hapus;
    TampilkanDataKamera(data, JumlahData);
    cout << "Pilih nomor kamera yang ingin dihapus: ";
    cin >> index;
    cin.ignore();
    index--;

    if (index >= 0 && index < JumlahData ){
        cout << "Yakin ingin hapus data kamera ini? (Y/N): ";
        cin >> konfir_hapus;

        if (konfir_hapus == 'Y' || konfir_hapus == 'y')
        {
            for (int i = index; i < JumlahData - 1; i++)
            {
                data[i] = data[i + 1];
            }
            JumlahData--;
                cout << "Data kamera berhasil dihapus!\n";
        }
        else
        {
            cout << "Penghapusan dibatalkan!\n";
        }
    }
    else
    {
        cout << "Nomor tidak valid!\n";
    }
}

int KembalikanKamera(int harga, int hari){
    if (hari == 0){
```

```
        return 0;
    }else{
        return harga + KembalikanKamera(harga, hari - 1);
    }
}

int KembalikanKamera(int harga, int lamaHari, float diskonSewa){
    float total = harga * lamaHari;
    float potongan = total * diskonSewa;
    cout << "Diskon: Rp" << potongan << endl;
    return total - potongan;
}

int main(){
    Kamera data[5];
    User user[2];
    int JumlahUser = 0;
    int JumlahData = 0;
    user[JumlahUser].username = "TWISTERSIAN";
    user[JumlahUser].password = "123Action";
    user[JumlahUser].role = "admin";
    JumlahUser++;
    bool MenuAwal = true;

    while(MenuAwal)
    {
        int PilihMenu;
        string InputNama, role_login;

        cout << "\n==========================\n";
        cout << "  HOME PAGE TWISTERS CAM\n";
        cout << "==========================\n";
        cout << "1. Register\n";
        cout << "2. Login\n";
        cout << "3. Keluar\n";
        cout << "Pilih (1-3): ";
        cin >> PilihMenu;
        cin.ignore();

        switch(PilihMenu)
        {
            case 1:
```

```
            RegisterUser(user, JumlahUser);
            break;

            case 2:
            if(!LoginUser(user, JumlahUser, InputNama, role_login)){
                return 0;
            }
            if (role_login == "admin")
            {
                int pilihan;
                do
                {
                    cout << "\n======================\n";
                    cout << "        MENU ADMIN\n";
                    cout << "======================\n";
                    cout << "1. Tambah Data Kamera\n";
                    cout << "2. Tampilkan Data Kamera\n";
                    cout << "3. Ubah Data Kamera\n";
                    cout << "4. Hapus Data Kamera\n";
                    cout << "5. Logout\n";
                    cout << "Pilih (1-5): ";
                    cin >> pilihan;
                    cin.ignore();

                    switch(pilihan)
                    {
                        case 1:
                        TambahDataKamera(data, JumlahData);
                        break;

                        case 2:
                        TampilkanDataKamera(data, JumlahData);
                        break;

                        case 3:
                        UbahDataKamera(data, JumlahData);
                        break;

                        case 4:
                        HapusDataKamera(data, JumlahData);
                        break;
                    }
```

```
                } while (pilihan != 5);
            }
            else
            {
                int pilihan;
                do
                {
                    cout << "\n======================\n";
                    cout << "        MENU USER\n";
                    cout << "======================\n";
                    cout << "1. Lihat Data Kamera\n";
                    cout << "2. Sewa Kamera\n";
                    cout << "3. Kembalikan Kamera\n";
                    cout << "4. Logout\n";
                    cout << "Pilih (1-4): ";
                    cin >> pilihan;

                    switch (pilihan)
                    {
                        case 1:
                        TampilkanDataKamera(data, JumlahData);
                        break;

                        case 2:
                        {
                            TampilkanDataKamera(data, JumlahData);
                            int pilih;
                            cout << "Pilih nomor kamera yang ingin
disewa: ";
                            cin >> pilih;
                            pilih--;

                            if (data[pilih].status.kondisi ==
"Tersedia")
                            {
                                cout << "Sewa berapa hari: ";
                                cin >> data[pilih].LamaSewa;
                                data[pilih].status.kondisi =
"Disewa";
                                data[pilih].NamaPenyewa = InputNama;
```

```
                                cout << "Kamera berhasil disewa!\n";
                            }
                            else
                            {
                                cout << "Kamera sedang disewa!\n";
                            }
                            break;
                        }

                        case 3:
                        {
                            TampilkanDataKamera(data, JumlahData);
                            int pilih;
                            cout << "Pilih nomor kamera yang Anda
sewa untuk dikembalikan: ";
                            cin >> pilih;
                            pilih--;
                            int total =
KembalikanKamera(data[pilih].HargaSewa, data[pilih].LamaSewa);
                            float pembayaran;

                            if(total >= 750000)
                            {
                                pembayaran =
KembalikanKamera(data[pilih].HargaSewa, data[pilih].LamaSewa, 0.15);
                                cout << "Harga sewa sebelum diskon:
Rp" << total << endl;
                            }
                            else
                            {
                                pembayaran = total;
                            }
                            cout << "Total harga sewa yang harus
dibayar: Rp" << pembayaran << endl;
                            data[pilih].status.kondisi = "Tersedia";
                            data[pilih].NamaPenyewa = "";
                            data[pilih].LamaSewa = 0;
                            cout << "Kamera berhasil dikembalikan\n";
                            break;
                        }
                    }
                } while (pilihan != 4);
```

```
            }
            break;

        case 3:
        cout << "Program selesai\n";
        return 0;

        default:
        cout << "Pilihan tidak valid\n";
        }
    }
    return 0;
}
```

4. Hasil Output

```
============================
  HOME PAGE TWISTERS CAM
============================
1. Register
2. Login
3. Keluar
Pilih (1-3):
```

```
============================
  HOME PAGE TWISTERS CAM
============================
1. Register
2. Login
3. Keluar
Pilih (1-3): 1

========================
      REGISTER USER
========================
Masukkan nama baru: Larry
Masukkan password baru: 068
Registrasi berhasil! Silakan login
```

```
========================
       MENU USER
========================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4):
```

```
====================
       LOGIN
====================
Masukkan username: TWISTERSIAN
Masukkan password: 123Action
Berhasil login! Selamat datang TWISTERSIAN
```

```
=========================
        MENU ADMIN
=========================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 
```

```
=======================
       MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 1

==========================================================================================
                          TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================================
ID Kamera: K01
Nama Kamera: NIKON 1974
Harga sewa/hari: Rp150000
Data berhasil ditambahkan

=======================
       MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 2

==========================================================================================
                           DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 1974          Rp150000            Tersedia
==========================================================================================
```

```
=======================
       MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 2

==========================================================================================
                           DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 1974          Rp150000            Tersedia
2     K02       SONY A7             Rp350000            Tersedia
==========================================================================================
```

```
=======================
        MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 3

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 1974          Rp150000            Tersedia
2     K02       SONY A7             Rp350000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin diubah: 2
ID Kamera baru: K02
Nama Kamera baru: SONY ALPHA
Harga sewa/hari: Rp450000
Status (Tersedia/Disewa): Tersedia
Data kamera berhasil diubah!
```

```
=======================
        MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 4

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 1974          Rp150000            Tersedia
2     K02       SONY ALPHA          Rp450000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin dihapus: 2
Yakin ingin hapus data kamera ini? (Y/N): Y
Data kamera berhasil dihapus!

=======================
        MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 2

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 1974          Rp150000            Tersedia
==========================================================================================
```

```
====================
       LOGIN
====================
Masukkan username: Larry
Masukkan password: 068
Berhasil login! Selamat datang Larry

=======================
       MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 1

==========================================================================================
                            DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NIKON 1974          Rp150000            Tersedia
==========================================================================================
```

```
=======================
       MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 2

==========================================================================================
                            DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NIKON 1974          Rp150000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin disewa: 1
Sewa berapa hari: 6
Kamera berhasil disewa!
```

```
=======================
       MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 3

==========================================================================================
                            DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NIKON 1974          Rp150000            Disewa
==========================================================================================
Pilih nomor kamera yang Anda sewa untuk dikembalikan: 1
Diskon: Rp135000
Harga sewa sebelum diskon: Rp900000
Total harga sewa yang harus dibayar: Rp765000
Kamera berhasil dikembalikan
```

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\praktikum-apl> git add "post-test/post-test-apl-3/2509106068-NATHANAEL LARRY GLASIA-PT-3.cpp"
PS C:\praktikum-apl> git add "post-test/post-test-apl-3/2509106068-NATHANAEL LARRY GLASIA-PT-3.exe"
```

Menambahkan file ke staging area untuk di commit

5.2 GIT Commit

```
PS C:\praktikum-apl> git commit -m "Menyelesaikan posttest 3 cuy"
[main f477513] Menyelesaikan posttest 3 cuy
 2 files changed, 358 insertions(+)
 create mode 100644 post-test/post-test-apl-3/2509106068-NATHANAEL LARRY GLASIA-PT-3.cpp
 create mode 100644 post-test/post-test-apl-3/2509106068-NATHANAEL LARRY GLASIA-PT-3.exe
```

Menyimpan perubahan yang ada di staging area ke repositori lokal dengan pesan

5.3 Git Push

```
PS C:\praktikum-apl> git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 81.68 KiB | 3.02 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/larrynathanx/praktikum-apl.git
   9fb1850..f477513  main -> main
```

Mengunggah perubahan dari repositori lokal ke repositori jarak jauh (GitHub)